

Designing Accessible Visualizations for People with Intellectual and Developmental Disabilities

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ABSTRACT

Visualization amplifies cognition and helps a viewer see the trends, patterns, and outliers in data. However, conventional visualization tools and guidelines do not actively consider the unique needs and abilities of people with Intellectual and Developmental Disabilities (IDD), leaving them excluded from data-driven activities and vulnerable to ethical issues in everyday life. My dissertation work explores the challenges and opportunities of cognitively accessible visualization. Through mixed-method approaches and close collaboration with people with IDD, my team and I ran experiments and developed guidelines to improve current visualizations, we interviewed people with IDD and gained initial understandings of their daily data experiences, and we are currently in the process of revising a participatory design workshop to create accessible visualizations for and with this population. For the remaining dissertation work, I hope to further expand our knowledge of cognitively accessible visualization, translating what I have learned from these experiences into a graphical user interface that supports people with IDD with their self-advocacy and self-expression using personally relevant data. My ultimate career goal is to theorize cognitively accessible visualization and empower people with IDD to make informed decisions and generate meaningful discoveries through accessible visual analytics.

CCS CONCEPTS

• **Human-centered computing** → **Visualization; Accessibility.**

KEYWORDS

Cognitive Accessibility, Personal Informatics, Physicalization, Storytelling

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1 INTRODUCTION

Data is everywhere, but it may not be accessible to everyone. Barriers to data access and use can create substantive inequities in society which lead to potentially devastating consequences for self-determination [5, 22], professional autonomy [21], and even basic healthcare [4]. Visualization aims to make data more useful and accessible by graphically representing complex data relationships and communicating data-driven insights. However, using visualizations requires a viewer to read abstract imagery, estimate statistics, and retain information. These processes usually function differently for people with Intellectual and Developmental Disabilities (IDD), such as Down syndrome, Williams syndrome, Fetal Alcohol syndrome, Autism, and Cerebral Palsy [27]. People with IDD typically face significant limitations in intellectual functioning and daily adaptive behaviors, which translate to challenges in reading traditional visualizations, and may require this population to experience and use data differently. However, these challenges and differences are seldom considered in conventional visualization tools and techniques, leaving people with IDD excluded from data-driven activities and vulnerable to many ethical issues in social participation.

My dissertation work aims to make data accessible to people with IDD, with a particular focus on designing accessible visualization to address real-world problems. The research was originally inspired by an ongoing study, *the State of the States in IDD* [5], and a partnership with the Coleman Institute for Cognitive Disabilities to design accessible visualizations to support the IDD community's financial self-advocacy and decision-making. In the first phase of my dissertation, I conducted a mixed-methods experiment to understand how people with and without IDD interpret data and common data visualizations. The results showed that people with IDD used different strategies to reason with data, and that common visualization guidelines may inevitably create barriers for them to access data [27]. These initial insights led me and my team to further investigate how people with IDD experience data in everyday life and to understand the role visualization plays in data accessibility. Through semi-structured interviews with 15 participants with IDD and their caregivers, we characterized the needs and challenges they face with data in everyday contexts, and identified opportunities for visualization to improve self-advocacy, self-determination, and self-expression. Following that, we are now preparing for a participatory design workshop to deeply understand this population, their ways of approaching data and creating visual representations. My short-term goal is to create an intuitive graphical user interface that supports accessible interactions between people with IDD and data; and my long-term objective is to establish cognitive visualization accessibility as a field of study, empowering individuals with varying cognitive abilities to enjoy working with data.

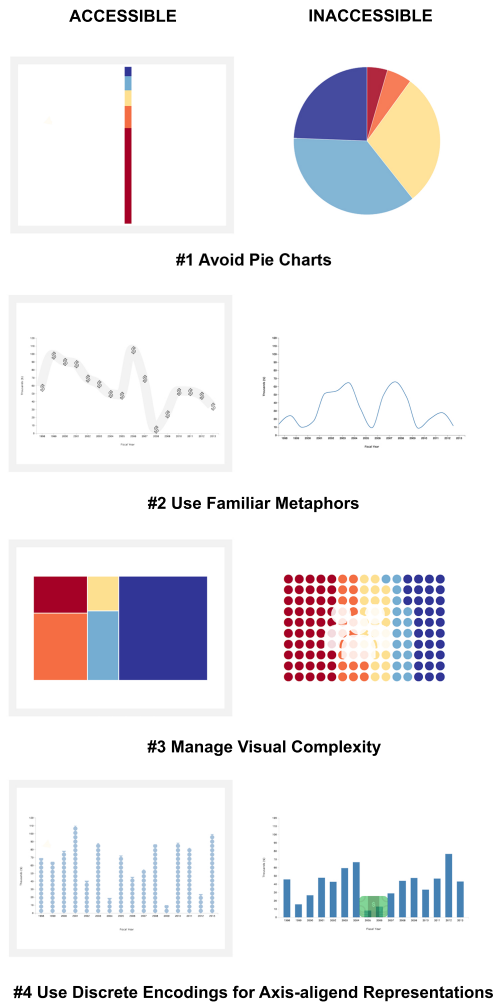


Figure 1: Accessible Visualization Design Guidelines Derived from the Experiment

2 BACKGROUND & RELATED WORK

Communicating Information to People with IDD. Intellectual disability (ID) is a developmental disability (DD) characterized by significant deficits in intellectual and adaptive functioning of varying severity presenting before 22 years of age [1]. The term IDD describes situations in which intellectual disability and other developmental disabilities are present, and it encompasses a broad spectrum of functioning and health conditions. Each individual with IDD has distinct strengths and weaknesses in their abilities but universally faces functional impairments in real-life skills, such as understanding rules, navigating daily living tasks, and participating in family, school, and community activities [2]. Many neurodevelopmental, psychiatric, and medical disorders co-occur with IDD, especially communication disorders, learning disabilities, epilepsy, and various genetically transmitted conditions. People with IDD also vary in their ability to understand written, spoken, pictorial, numerical, or sign language and to express themselves through

these diverse media [10]. Most people with IDD use speech or sign as their predominant form of communication [10] and primarily receive information verbally [24]. About 60% of people with IDD are able to use symbolic methods such as pictures, symbols, signs or speech to communicate [8]. Some people lacking verbal abilities may be able to respond to pictorial narrative [3]. Those who are nonverbal and only use pre-symbolic communication largely rely on the people around them, such as peers and caregivers, to anticipate their needs and to interpret their vocalizations, facial expressions, and body language [20]. These cognitive and communicative differences make creating accessible information for people with IDD highly complex, as accessible solutions must address diverse individual abilities, communication strategies, and preferences. Conventional data visualization tools and guidelines are primarily designed for neurotypical consumers to solve domain-specific problems with uniform solutions. They rarely capture the diverse needs and abilities of people with IDD, leaving this population with limited access to data important to their well-being. This dissertation work aims to understand how visualization can better accommodate the needs of neurodiverse audiences, and we adopt a human-centered approach to designing solutions grounded in the authentic lived experiences of people with IDD.

Cognitive Accessibility & Inclusive Visualization. Cognitive and learning disabilities can affect people’s perception, memory, language, attention, problem solving, and comprehension. WCAG gives advice on how to make web content and applications accessible for this user group [12]. They suggest using clear and understandable content, helping users focus and find what they need, supporting adaptation and personalization, and avoiding rely on memory to navigate the web experience. However, these guidelines treat information processing at a high level, and can not be applied directly to the design of data visualization, as data usually comes in a variety of formats and requires more nuanced guidelines to reflect the relationship and structure. On the other hand, carefully crafted interactive visualizations can reach broad audiences [14], and have the potential to empower people from all walks of life, including those with a range of abilities [15, 18], when designed with attention to the target user’s needs. Past work in accessible visualization has illustrated barriers to access [13, 27] as well as methods [6, 16] and techniques [11, 17, 28, 30] for fostering data access. While visualization accessibility is gaining increasing attention, the primary focus of past research has been on visual accessibility. However, physical and cognitive abilities can also influence how people work with data visualizations [18]. For example, past studies have investigated how particular designs may induce seizures in people with epilepsy [7, 23]. Inspired by the State of the States in IDD initiative [5], we conducted a mixed-methods experiment to understand how people with IDD interpret common data visualizations [27]. The experiment revealed that conventional visualization guidelines may inadvertently create barriers to data use and provided insight into the reasoning strategies and visualization preferences of people with IDD. Drawing on insights and momentum from that study, my dissertation work will continue focusing on cognitive accessibility, and further investigate the role of visualization in improving data accessibility.

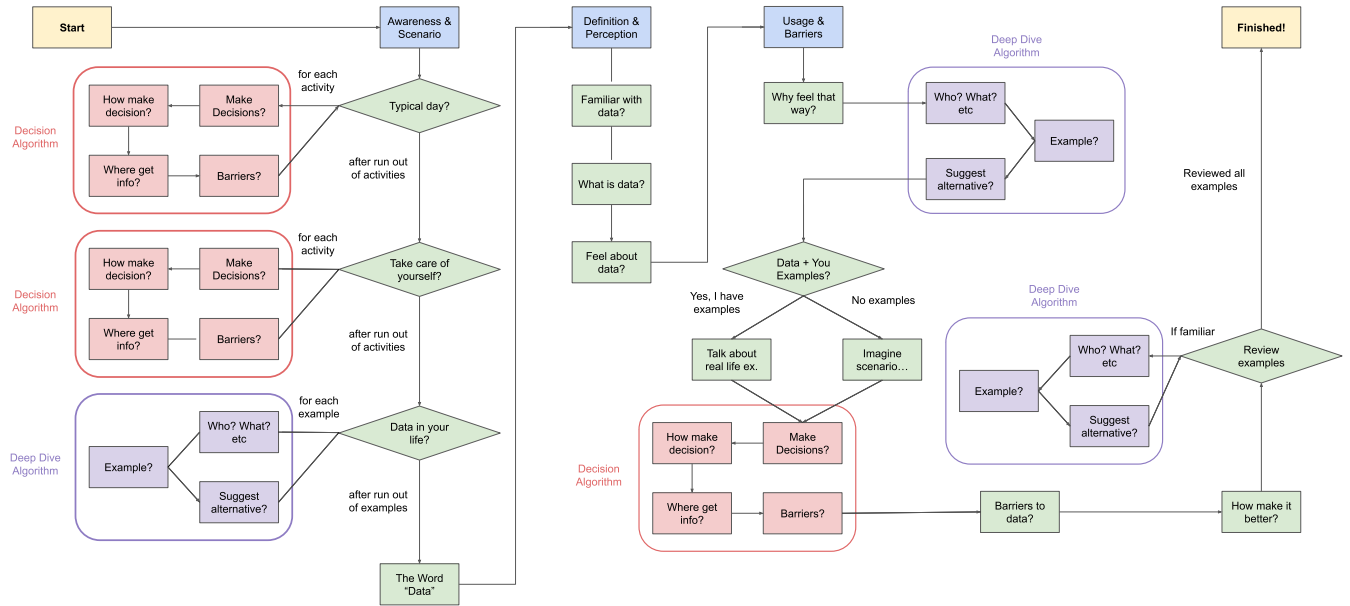


Figure 2: Flowchart of the Algorithmic Interview Process

3 RESEARCH QUESTIONS AND METHODS

My dissertation will address three primary research questions: (1) how people with IDD interpret common visualizations, (2) how people with IDD encounter data in everyday life, and (3) how people with IDD approach data and create visualizations. We follow a human-centered approach and use a combination of quantitative and qualitative methods.

3.1 How Do People With IDD Interpret Common Visualizations?

Literature on educational psychology [26, 29], mathematical studies [19, 25], and graphical perception [9, 29] suggests that people with IDD use different strategies to reason about abstract information. Building on collaborative efforts with domain experts [5], we identified three design elements that may improve data accessibility—*chart type*, *chart embellishment*, *data continuity*—and made three hypotheses that

H1- The best chart type for a given task will differ between people with and without IDD.

H2- Semantically meaningful chart embellishments will enhance data interpretation for people with IDD.

H3- Discrete data representations will lead to more accurate performance for people with IDD.

Methods: To test our hypotheses and understand how each of the design elements may influence accessibility, we developed a **web-based mixed-methods experiment**, and rendered the stimuli using time-series data and proportion data in the context of financial data analysis and demographic information interpretation. For the quantitative part, we collected participants' task performance time and accuracy; for the qualitative part, we asked participants to talk

out loud their chart preferences and the strategies they used when reasoning with data. We additionally recruited people without IDD as a control group to better understand the differences in visual perception, reasoning and sense-making.

Progress: Completed. Findings were summarized as 4 design guidelines, detailed in [27] and illustrated in Fig.1.

3.2 How Do People With IDD Encounter Data In Everyday Life?

To design visualizations that benefit people with IDD in real life, we need to understand the relationship between people with IDD and data in everyday contexts. Such understanding can lead to solutions and novel visualizations that provide concrete support in independent living and critical areas of social participation. Particularly, we need to situate data in the lived experiences of people with IDD and characterize

- (1) How they conceptualize data;
- (2) When and where they encounter data;
- (3) What barriers exist when they interact with data.

Methods: Given the exploratory nature of this question, qualitative methods, such as **semi-structured interview** would suit the best. However, since abstract thinking and verbal communication are two prominent challenges facing people with IDD, the interview should be open-ended and flexible. We therefore adopted an algorithmic approach to actively uncover otherwise hidden data encounters in the everyday life of people with IDD while also acknowledge their agency. The interview flow is illustrated in Fig.2

Progress: Completed and Under Review at CHI'23. The interview revealed that people with IDD frequently used personal data to



Figure 3: Conceptual Figure of the Workshop

make sense of everyday life, and highlighted the needs of making data feel personal and designing more approachable visualizations.

3.3 How Do People With IDD Approach Data And Create Visualizations?

Truly accessible visualization should include people with disabilities to design solutions to addressing their most pressing problems. To promote engagement and authenticity, the tasks, data-sets, and procedures need to respect individual differences and reflect participants' interests. The study needs to be a pleasant and personally relevant experience. We adopt participatory design approaches and are particularly interested in understanding

Q1 How people with IDD approach different types of data.

Q2 How visualization design can link data to positive identity.

Q3 How visualizations can support self-expression through data.

Methods: We invite people with IDD to a 6-week remote **participatory design workshop** to create visualizations and foster visual communication. The activities are designed in the context of teaching aliens from the VisualLand about different aspects of Earth. Four aliens— Foodie, Athlete, Artist, and Gamer—correlates with four topics that are used in the workshop to assign datasets and find shared interests. The workshop concept is shown in Fig.3. Participants are paired up as “alien buddies” and give each other asynchronous feedback every week.

Progress: In progress. So far, 4 participants have completed the 6-week activities. However, we encountered several issues such as retaining participants, accommodating different types and levels of disabilities, reoccurring task confusion and technological frustration etc. We are currently revisiting the design and would benefit from feedback from the DC as to the best path forward to answer this research question.

4 NEXT STEPS

Building on these progresses, I hope to prototype an intelligent system that helps people with IDD with accessible data analytics, and ultimately theorize cognitively accessible visualization. I am interested in incorporating visual aesthetics principles and storytelling techniques to elicit actionable reflection in personal data and generate positive emotional responses in the viewer using and authoring data visualizations.

5 DISSERTATION STATUS AND LONG TERM GOALS

I am in my fourth year and plan to graduate in Spring 2024. The short-term goal is to translate knowledge I have gained in the last few years into a functional prototype that assists people with IDD to work with data. The long-term goal is to lay out a concrete road-map for cognitively accessible visualization, and establishes it as a field of study for future investigation.

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